

# Daedalus Aero-Structural Design Report

## Fixed-Wing Remote Controlled Aircraft

August 14, 2026

## 1 Baseline Design & Configuration Choices

This report documents the aerodynamic layout, structural sizing, and manufacturing decisions for a lightweight 1.0 m wingspan fixed-wing trainer RC aircraft.

### 1.1 Aircraft Parameter Summary

Table 1: Aircraft Parameter Summary

| Parameter                            | Value                             | Rationale                                                                                                          |
|--------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| All-Up Weight                        | 1.0 kg ( $\approx 9.81$ N)        | Target operating weight for 3D-printed RC trainer class                                                            |
| Design Load Factor ( $n$ )           | 3g maneuver limit                 | Accommodates maximum pull-up stresses (29.43 N total lift)                                                         |
| Wingspan ( $b$ )                     | 1000 mm (1.0 m)                   | Optimized for 3D printing and easy vehicle transport, whilst large enough for smooth flying and visibility         |
| Root Chord ( $C_r$ )                 | 180 mm                            | Allow for internal spar clearance and structural strength at wing root                                             |
| Tip Chord ( $C_t$ )                  | 120 mm                            | Taper ratio $\lambda = 0.67$ to approximate elliptical lift distribution                                           |
| Mean Aerodynamic Chord ( $\bar{c}$ ) | 150 mm                            | Total Wing Area $S = 0.15$ m <sup>2</sup> (Aspect Ratio $AR = 6.67$ )                                              |
| Aerofoil Profile                     | NACA 2412                         | High lift-to-drag ratio, gentle stall characteristics, clean trailing edge                                         |
| Wing Planform                        | Straight Trailing Edge (0° Sweep) | Flat alignment datum for workbench assembly; straight trailing edge allows for easier aileron attachment           |
| Dihedral Angle                       | 0° (Flat Wing)                    | Enables continuous straight spar pass-through without angled joiners                                               |
| Main Spar Sizing                     | 6 mm OD / 4 mm ID Carbon Tube     | Single continuous 750 mm carbon fiber tube calculated to withstand Aerodynamic forces with a high factor of safety |

## 1.2 Key Architectural Decisions

### A. Zero Dihedral (Flat Wing Architecture)

- **Manufacturing Efficiency:** Eliminates complex 3D-printed angled structural joiners at the root of the wing. A single continuous 750 mm carbon tube passes directly through wing panels without structural discontinuities.
- **Stability Recovery:** Roll stability lost by removing dihedral is compensated by the **high-wing mounting configuration**, exploiting natural pendulum stability (keeping the CG well below the aerodynamic center of lift).
- **Crash Resilience:** Zero dihedral does make a wing tip strike more likely during takeoff and landing as the wing tip is closer to the ground; however, again this is compensated with the high wing arrangement and the Elastic Wing Attachment System.

### B. Flat Trailing Edge Reference Point

- **Assembly Datum:** Keeping the trailing edge orthogonal to the fuselage provides a straight reference line for alignment during manufacturing and provides a straight section for the carbon spar to sit.
- **Spar Clearance:** Placing a straight spar parallel to the flat trailing edge maintains printed wall clearance top and bottom across the entire span.
- **Aileron Placement:** Keeping a flat trailing edge allows for more roll maneuverability as the ailerons can be mounted orthogonal to the direction of airspeed.

### C. Elastic Wing Attachment System

- **Crash Resilience:** The proposed idea is to use elastic bands to attach the wing to the fuselage. During a wing strike, the impact will be partially absorbed by the elastic and during a violent crash, the wing may detach, all to protect the fuselage and its contents.
- **Redundancy:** The design should allow for mechanical attachment (screws/bolts) in the event that testing shows too much flex from the rubber band attachment system impacting flight dynamics.

## 2 Structural Analysis: Spar Sizing (Euler-Bernoulli Beam)

To verify that reducing the main carbon spar diameter from 8 mm to 6 mm satisfies structural safety limits under a **3g pull-up maneuver**, maximum bending stress ( $\sigma_{max}$ ) was computed using Euler-Bernoulli beam theory.

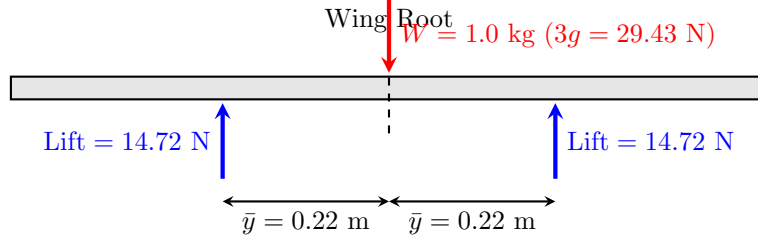


Figure 1: Free body diagram for 3g pull-up maneuver beam loading.

### 2.1 A. Load Condition & Root Bending Moment

For a 3g maneuver on a 1.0 kg aircraft, total required aerodynamic lift is:

$$L_{total} = n \cdot W = 3 \times 9.81 \text{ N} = 29.43 \text{ N} \quad (1)$$

Each semi-span wing panel carries half the load (14.72 N). For a tapered planform ( $\lambda = 0.67$ ), the spanwise lift centroid sits at approximately  $y_{bar} = 0.22 \text{ m}$  from the wing root:

$$M_{root} = L_{panel} \cdot y_{bar} = 14.72 \text{ N} \times 0.22 \text{ m} = 3.24 \text{ N} \cdot \text{m} \quad (2)$$

### 2.2 B. Second Moment of Area ( $I$ )

For a hollow circular pultruded carbon fiber tube with outer diameter  $D_{outer} = 0.006 \text{ m}$  (6 mm) and inner diameter  $D_{inner} = 0.004 \text{ m}$  (4 mm):

$$I = \frac{\pi}{64} (D_{outer}^4 - D_{inner}^4) = \frac{\pi}{64} ((0.006)^4 - (0.004)^4) \approx 5.105 \times 10^{-11} \text{ m}^4 \quad (3)$$

### 2.3 C. Bending Stress Calculation

The maximum distance from the neutral axis to the outer fiber is  $y_{max} = \frac{D_{outer}}{2} = 0.003 \text{ m}$ :

$$\sigma_{max} = \frac{M_{root} \cdot y_{max}}{I} = \frac{3.24 \text{ N} \cdot \text{m} \times 0.003 \text{ m}}{5.105 \times 10^{-11} \text{ m}^4} \approx 190.4 \text{ MPa} \quad (4)$$

### 2.4 D. Safety Factor Evaluation

Pultruded carbon fiber tubing exhibits a typical ultimate flexural strength of  $\sigma_{ult} \approx 1200 \text{ MPa} - 1500 \text{ MPa}$ :

$$\text{Safety Factor (SF)} = \frac{\sigma_{ult}}{\sigma_{max}} = \frac{1200 \text{ MPa}}{190.4 \text{ MPa}} \approx \mathbf{6.3} \quad (5)$$

### 3 Aerodynamic Simulation & Incidence Optimization (VSPAERO)

Aerodynamic performance was evaluated using Vortex Lattice Method (VLM) solver passes in **OpenVSP** / **VSPAERO**.

### Simulation Parameters

- **Flight Velocity ( $V$ ):** 15 m/s (Mach = 0.044 at sea level)
- **Reynolds Number ( $Re$ ):**  $1.5 \times 10^5$  (based on 150 mm MAC)
- **Angle of Attack Sweep ( $\alpha$ ):**  $-2.0^\circ$  to  $+8.0^\circ$

### 3.1 A. Cruise Target Determination

At level cruise (15 m/s, dynamic pressure  $q = \frac{1}{2}\rho V^2 = 137.8$  Pa), the required target lift coefficient ( $C_{L,cruise}$ ) to support 1.0 kg is:

$$C_{L,target} = \frac{W}{a \cdot S} = \frac{9.81 \text{ N}}{137.8 \text{ Pa} \times 0.15 \text{ m}^2} \approx \mathbf{0.475} \quad (6)$$

### 3.2 B. Angle of Incidence Adjustment

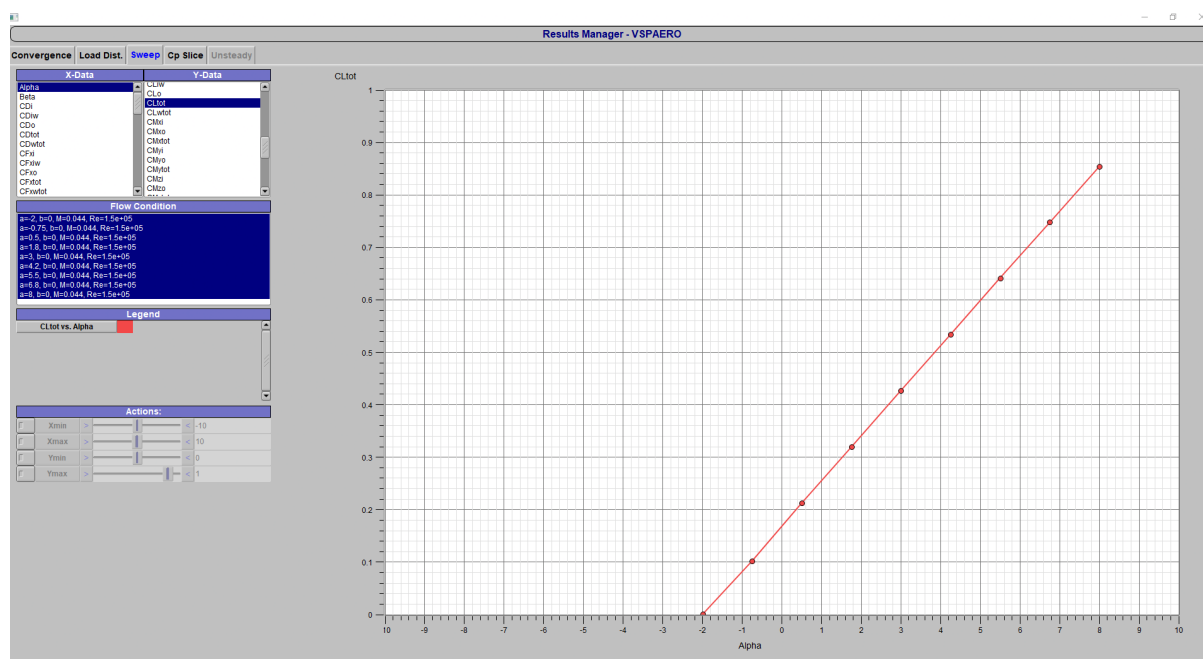


Figure 2: Initial wing sweeps showed that an un-pitched NACA 2412 section produced  $C_L \approx 0.18$  at  $\alpha = -2^\circ$  and required a fuselage pitch of  $\alpha \approx +3.8^\circ$  to generate  $C_L = 0.475$ .

**Original: High Fuselage Drag**  
+3.8° Fuselage Pitch (Nose Up)

**Adjusted: Level Fuselage Flight**  
0° Fuselage Pitch / Level Thrust

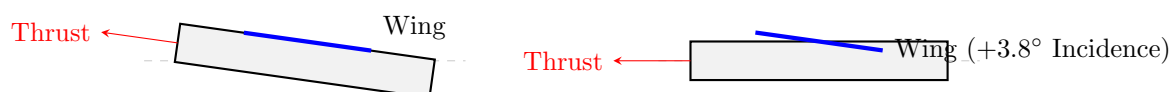


Figure 3: Comparison of fuselage attitude and thrust vector alignment during level cruise.

To optimize level-flight efficiency, the wing mounting saddle was designed with a built-in  $+3.8^\circ$  **incidence angle** relative to the fuselage center axis and motor thrust line:

1. **Drag Reduction:** The primary fuselage body sits at  $\alpha_{body} \approx 0^\circ$  during normal cruise, minimizing frontal parasitic drag ( $C_{D0}$ ).
2. **Trim Optimization:** Aligns the thrust vector directly along the direction of flight, avoiding power pitch coupling during level cruise.

### 3.3 C. Spanwise Load Distribution Validation

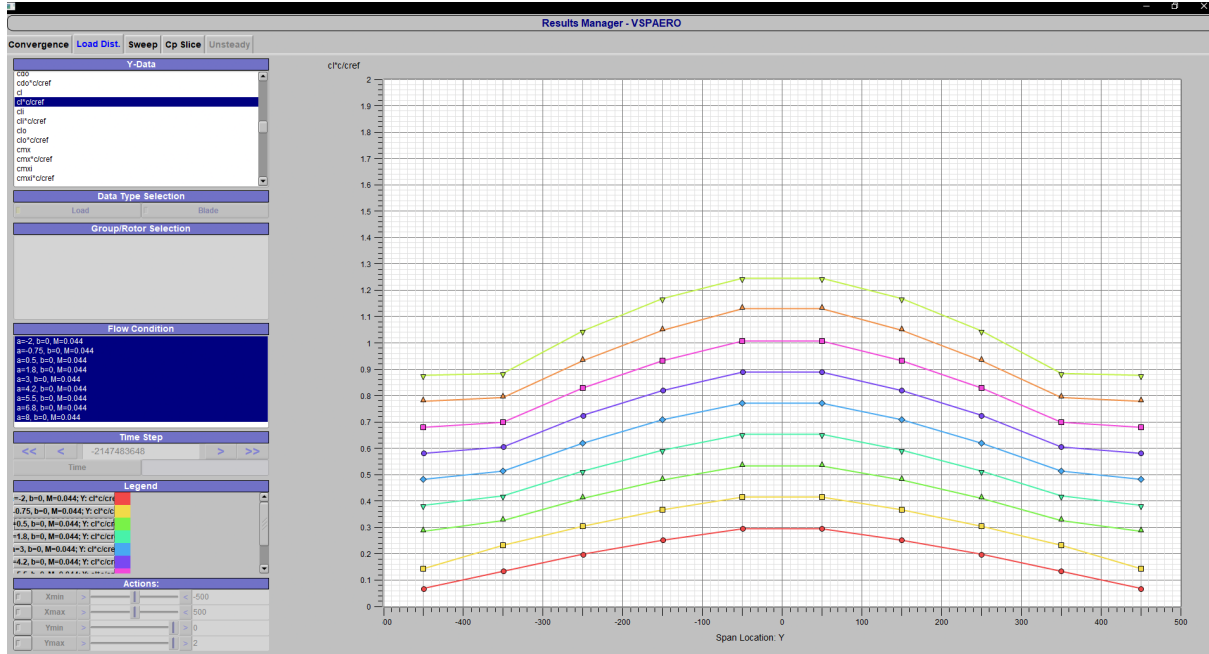


Figure 4: VSPAERO output for spanwise circulation ( $c_l \cdot c / c_{ref}$ ) confirmed that the chosen 180 mm  $\rightarrow$  120 mm planform taper successfully rounds the lift curve toward an elliptical distribution. This minimizes induced drag ( $C_{Di}$ ) and prevents localized premature tip stall without needing complex geometric twist (washed-out tips).

## 4 Empennage Design & Static Stability Analysis

### 4.1 Tail Surface Geometry & Sizing

The empennage was sized to provide pitch and directional stability for a high-wing RC trainer configuration ( $S = 0.15 \text{ m}^2$ ,  $\bar{c} = 0.15 \text{ m}$ ,  $b = 1.0 \text{ m}$ ). To simplify manufacturing, both the horizontal stabilizer and vertical fin utilize a uniform taper ratio ( $\lambda = 0.67$ ) with a straight  $0^\circ$  trailing edge sweep. This maintains a straight hinge line for control surface actuation.

To accommodate a single continuous elevator across the full horizontal span (320 mm), the vertical fin is shifted 25 mm forward relative to the horizontal tail plane ( $X_{fin} = 445 \text{ mm}$ ,  $X_{HT} = 470 \text{ mm}$ ).

Table 2: Empennage Geometric Parameters and Tail Volume Coefficients

| Parameter                    | Horizontal Tail (HT)                       | Vertical Fin (VT)                              |
|------------------------------|--------------------------------------------|------------------------------------------------|
| Root Chord ( $C_r$ )         | 120 mm                                     | 105 mm                                         |
| Tip Chord ( $C_t$ )          | 80 mm                                      | 70 mm                                          |
| Span / Height ( $b_h/h_v$ )  | 320 mm                                     | 150 mm                                         |
| Surface Area ( $S_t$ )       | $0.032 \text{ m}^2$ ( $320 \text{ cm}^2$ ) | $0.0131 \text{ m}^2$ ( $131.25 \text{ cm}^2$ ) |
| Moment Arm ( $l_t$ )         | 0.450 m                                    | 0.450 m                                        |
| Incidence Angle ( $i_t$ )    | $-1.0^\circ$                               | $0.0^\circ$                                    |
| Volume Coefficient ( $V_t$ ) | <b><math>V_h = 0.64</math></b>             | <b><math>V_v = 0.039</math></b>                |

The resulting volume coefficients fall squarely within standard trainer aircraft stability boundaries:

$$V_h = \frac{S_h \cdot l_h}{S \cdot \bar{c}} = \frac{0.032 \cdot 0.45}{0.15 \cdot 0.15} = 0.64 \quad (0.50 - 0.70 \text{ target range}) \quad (7)$$

$$V_v = \frac{S_v \cdot l_v}{S \cdot b} = \frac{0.0131 \cdot 0.45}{0.15 \cdot 1.0} = 0.039 \quad (0.035 - 0.045 \text{ target range}) \quad (8)$$

### 4.2 Control Surface Dimensioning

Control surfaces were dimensioned relative to local surface chords to deliver responsive handling while maintaining forgiving control characteristics for a trainer aircraft:

- **Ailerons:** 20% wing chord ratio (30 mm chord) positioned outboard from  $Y = 250 \text{ mm}$  to  $Y = 475 \text{ mm}$  (45% of semi-span). A fixed 25 mm wingtip cap (5% semi-span) is retained to protect moving corners from ground contact during landing. The hinge line sits 120 mm aft of the wing leading edge (80% chord).
- **Elevator:** 25% horizontal stabilizer chord ratio (30 mm root, 20 mm tip). The single continuous surface spans 310 mm (5 mm fixed protective tip caps on outer ends) with a 33 – 35 mm overhang clear of the rear fuselage.
- **Rudder:** 35% vertical fin chord ratio (37 mm root, 25 mm tip) spanning the full 150 mm fin height to ensure crisp yaw authority and ground handling.

Table 3: Primary Control Surface Sizing Breakdown

| Surface       | Chord Ratio | Root / Tip Chord | Spanwise Extent                        | Deflection Range                        |
|---------------|-------------|------------------|----------------------------------------|-----------------------------------------|
| Ailerons (x2) | 20%         | 30 mm/30 mm      | $Y = 250 \text{ to } 475 \text{ mm}$   | $\pm 12^\circ \text{ to } \pm 15^\circ$ |
| Elevator      | 25%         | 30 mm/20 mm      | $Y = -155 \text{ to } +155 \text{ mm}$ | $\pm 15^\circ \text{ to } \pm 20^\circ$ |
| Rudder        | 35%         | 37 mm/25 mm      | $Z = 15 \text{ to } 165 \text{ mm}$    | $\pm 20^\circ \text{ to } \pm 25^\circ$ |

### 4.3 Longitudinal Static Stability Verification ( $C_m$ vs. $\alpha$ )

A Vortex Lattice Method (VLM) aerodynamic simulation was executed in OpenVSP/VSPAERO at  $V = 15$  m/s across an angle-of-attack sweep ( $\alpha = -2^\circ$  to  $+8^\circ$ ).

The aircraft Center of Gravity was set to  $X_{cg} = 226.5$  mm (46.5 mm behind the wing leading edge, corresponding to 31% MAC), yielding a target Static Margin ( $SM$ ) of 10%:

$$SM = \frac{X_{np} - X_{cg}}{\bar{c}} = \frac{241.5 \text{ mm} - 226.5 \text{ mm}}{150 \text{ mm}} = 0.10 \quad (9)$$

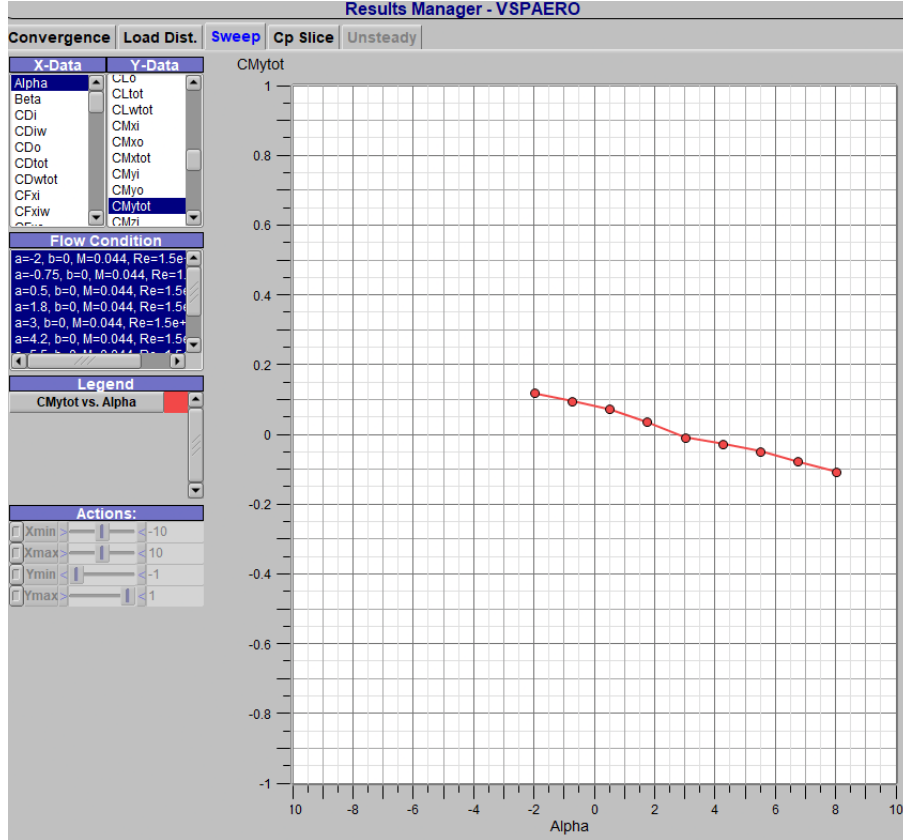


Figure 5: Pitching moment coefficient ( $C_{M,tot}$ ) versus angle of attack ( $\alpha$ ) showing negative static pitch slope and hands-off level trim at  $\alpha \approx 3.0^\circ$ .

The VSPAERO pitching moment curve demonstrates strong static longitudinal stability:

- **Negative Stiffness Slope:**  $\frac{dC_m}{d\alpha} < 0$  across the full operational alpha range, confirming natural pitch restoration following atmospheric perturbations.
- **Trim Point:** Neutral pitch trim ( $C_m = 0$ ) occurs at a body pitch angle of  $\alpha \approx +3.0^\circ$ , matching the wing's mounted incidence angle ( $+3.8^\circ$ ) to provide stable hands-off cruise trim.
- **Tail Downforce:** Spanwise lift distributions confirm the horizontal tail generates negative lift ( $c_l \cdot c/c_{ref} < 0$ ), balancing the wing's nose-down pitching moment ( $C_{m0}$ ).

## 5 Detailed CAD Integration, Internal Layout & Manufacturing Optimization

Following aerodynamic verification, extensive detail work was conducted in Autodesk Fusion to transform the outer aerodynamic shell into a manufacturable, 3D-printable airframe with integrated internal bays and simplified assembly features.

### 5.1 Internal Fuselage Architecture and Manufacturing Features

To ensure ease of 3D printing and rapid electronics assembly, significant focus was placed on internal volume planning and structural integration:

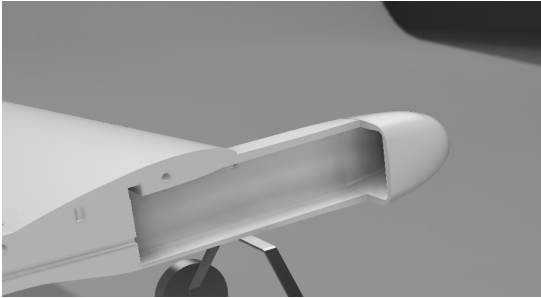


Figure 6: The primary fuselage is modeled with a large, hollow central equipment bay to house the 3S LiPo battery, ESC, ExpressLRS receiver, and other potential future electronics. This design maximizes interior airflow for thermal management while providing a flexible battery sliding range to adjust the CG during final balance testing. See the cable channel coming from the rear tail servos in this section view.

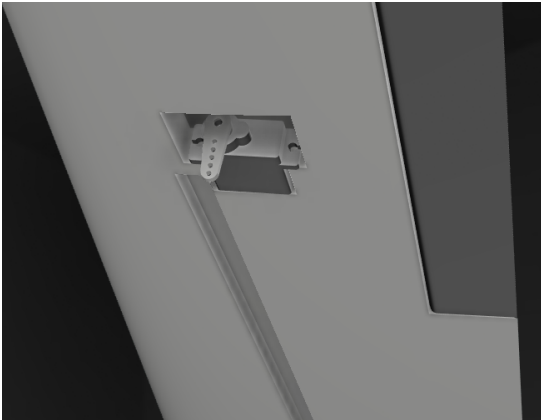


Figure 7: **Dedicated Servo Bay Enclosures:** Pocket recesses and internal mounting frames were modeled directly into the wing undersides and rear tail boom. These pockets allow 9g metal-gear servos to drop flush into the surface, eliminating external aerodynamic drag and providing rigid anchor points. Wire push-through channels route servo leads internally back to the main receiver bay.

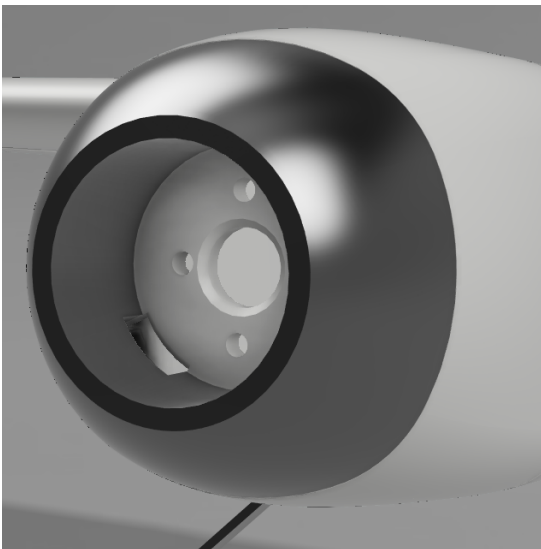


Figure 8: **Motor Firewall & Nose Cowl System:** The nose incorporates a thick internal motor mounting firewall with modeled bolt patterns for 28 mm outer-diameter brushless motors (XXD 2212), and cooling vents which double as cable routing channels for the motor cable. A detachable aerodynamic nose cowl fits over the motor assembly, shielding wiring while providing easy access for maintenance. The entire nose section is removeable too to allow easy access to the back of the firewall for motor mounting/maintenance. The nose is modelled with a  $2^\circ$  rotation right yaw and  $2^\circ$  rotation down pitch to counter motor torque, spiraling slipstream yaw and throttle induced pitch.



## 5.2 Landing Gear Positioning and Takeoff and Landing Stance

The positioning of the landing gear and tail wheel was carefully calculated to ensure ground clearance from the propeller, and to optimise the attitude of the plane for easier take off and landings in rough terrain.

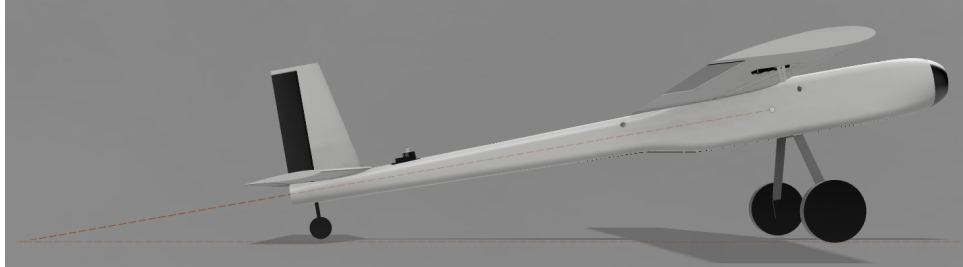


Figure 9: CAD ground line alignment establishing a  $+10^\circ$  nose-up attitude ( $\alpha_{ground}$ ) and main axle forward displacement relative to the Center of Gravity.

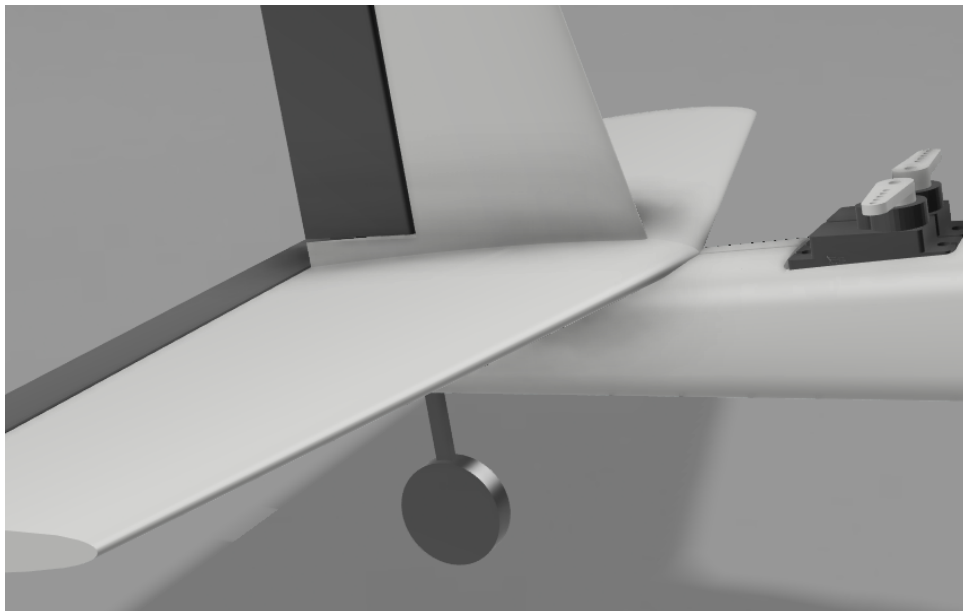


Figure 10: Optimized tail wheel/skid location situated at the extreme aft tail cone directly under the rudder line to minimize bending leverage on the printed fuselage.

- **Main Axle Position:** Placed directly under the wing leading edge (46.5 mm ahead of  $X_{cg}$ ), following the  $15^\circ - 20^\circ$  forward angle rule to prevent tipping during throttle spool-up.
- **Ground Stance Pitch** ( $\alpha_{ground} = 10^\circ$ ): Combines 65 mm main wheels on a 70 mm aluminum strut with the aft tail wheel, allowing the wing to build lift naturally during takeoff roll.
- **Structural Load Path:** The rear wheel mount was shifted from the mid-fuselage to the extreme aft tail cone (Figure 10), transferring vertical landing shocks directly into the bulkheads rather than flexing the tail boom.

### 5.3 Bill of Materials (BOM) and Cost Analysis

To establish a complete expenditure baseline, all airframe components, electronic hardware, raw materials, and power systems were itemized. Manufacturing is based on an FDM 3D printer in-house, with a structural print mass estimated at approximately 800 g of PLA. Currently projected to be under our £100 budget.

Where existing equipment is utilized—such as the radio transmitter and steel pushrod wire—costs are recorded as £0.00 to distinguish direct new project expenditure from existing inventory.

Table 4: Airframe Hardware, Avionics, and Materials Bill of Materials (BOM)

| Component Description                 | Qty / Mass     | Cost (£)      | Selection / Sourcing Rationale                                                                 |
|---------------------------------------|----------------|---------------|------------------------------------------------------------------------------------------------|
| <b>PLA Filament (1.75mm)</b>          | 1 Spool (1 kg) | £11.99        | In-house 3D printing; $\approx$ 800 g estimated print mass for airframe and support structures |
| <b>3S LiPo Battery (11.1V)</b>        | 1              | £13.00        | Sourced via Hobby RC (local UK stock alternative to AliExpress)                                |
| <b>XXD A2212 1000KV Motor</b>         | 1              | £4.40         | Proven trainer thrust profile on 3S                                                            |
| <b>Hobbywing Skywalker 40A ESC</b>    | 1              | £13.59        | Integrated 5V 3A SBEC prevents receiver brownouts                                              |
| <b>MG90D Metal-Gear Servos</b>        | 4              | £9.60         | Metal gears prevent shock damage during grass landings                                         |
| <b>HappyModel EPW6 ELRS Receiver</b>  | 1              | £11.39        | Native PWM output; includes dipole antenna out of the box                                      |
| <b>APC 9x6 Precision Propellers</b>   | 1 (2-Pack)     | £3.06         | Dynamic thrust match for 1000KV motor                                                          |
| <b>6mm OD Carbon Fiber Spars</b>      | 2 (500 mm)     | £8.69         | Primary wing bending reinforcement ( $SF = 6.3$ )                                              |
| <b>Aluminum Gear + 65mm Wheels</b>    | 1 Set          | £5.65         | Cushions rough field landing impacts                                                           |
| <b>Tail Wheel Assembly</b>            | 1              | £2.02         | Light aft tail cone support                                                                    |
| <b>CA Control Surface Hinges</b>      | 1 Pack         | £1.80         | Flexible paper/nylon CA hinges for control surfaces                                            |
| <b>3D-Printed Control Horns</b>       | 4              | £0.00         | Modeled in CAD; printed directly using main PLA spool                                          |
| <b>Pushrods (Piano Wire)</b>          | 1 Set          | £0.00         | Existing inventory / owned hardware                                                            |
| <b>RadioMaster Pocket Transmitter</b> | 1              | £0.00         | Existing inventory / owned EdgeTX transmitter                                                  |
| <b>Total New Project Outlay</b>       | —              | <b>£85.19</b> | <b>Direct component and material purchasing cost</b>                                           |

## 5.4 Measurement-Driven CAD Next Steps

To ensure sub-millimeter fitment and avoid printed slot misalignment, remaining mounting cuts have been deliberately deferred:

1. **Control Horn Mounting Slots:** Slots in the ailerons, elevator, and rudder will be cut in CAD after physical delivery of control horns to measure horn baseplate pin spacing with calipers.
2. **Landing Gear Floor Mount:** Mounting bolt holes in the internal 3.5 mm fuselage reinforcement plate will be drilled/cut based on direct physical measurement of the aluminum bracket hole spacing.

## 6 Future Work & Post-Maiden Flight Optimization

Initial flight testing (Maiden Flight) will be conducted using a 1-to-1 direct stick control layout (Channel 1: Left Aileron, Channel 2: Elevator, Channel 3: Throttle, Channel 4: Rudder, Channel 5: Right Aileron).

Following baseline flight validation and physical trim adjustments, radio software optimization will be implemented in the RadioMaster Pocket (EdgeTX):

- **Mechanical & Digital Sub-Trim:** Fine-tuning output center offsets in EdgeTX so control surfaces sit perfectly flush with aerofoil trailing edges during level cruise.
- **Electronic Aileron Differential:** Programming reduced downward aileron deflection relative to upward deflection (e.g., 100% UP / 60% DOWN) to suppress adverse yaw drag during turns.
- **Flaperon Mixing:** Configuring a 2-position transmitter switch to drop both ailerons downward simultaneously by  $10^\circ - 15^\circ$  during approach, acting as landing flaps to slow descent speed on short fields.
- **Future Payloads:** Once the aircraft has been trimmed and optimised, future projects may include adding a sensor suite to take data such as attitude, true air speed, altitude etc. Potentially even programming an automated aerial maneuver.

to be continued...